TEST PLAN AND EXECUTION EVIDENCE

Test ID	Module	Purpose	Execution result
TC01	Port & Access Audit	Run bounded Nmap scan against loopback target.	Executed; 127.0.0.1 up; 8080/tcp open; 99 top-100 ports closed.
TC02	Port & Access Audit	Check HTTP rate-limit indicators using five requests.	Executed; all five returned 404; assessment inconclusive.
TC03	AI Threat Detection	Upload and classify labelled flow CSV.	Executed; 6,000 rows processed; 99.38% agreement with labels.
TC04	Security Report	Generate consolidated PDF.	Executed; report generated and reviewed.

TC01 — Bounded Nmap scan

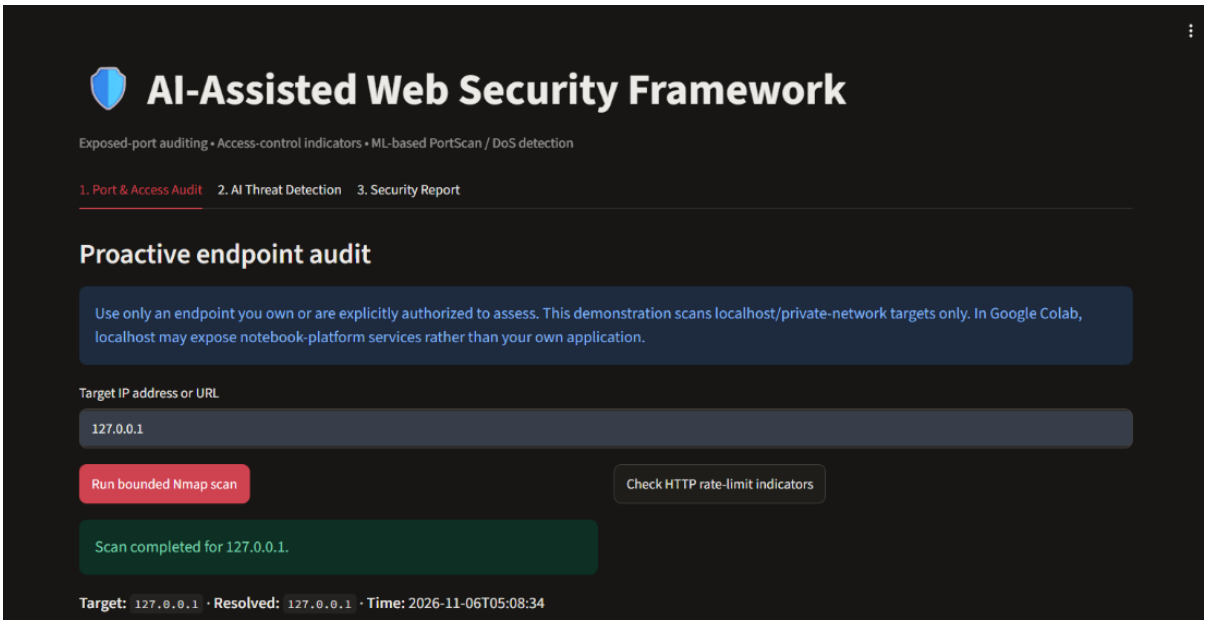
Field	Details
Objective	Verify that the application can run a bounded Nmap scan and display open TCP ports.
Test input	127.0.0.1 (loopback target inside Google Colab).
Expected outcome	The target is checked within the configured scope; open TCP ports and service details are displayed.
Actual result	Nmap 7.94SVN reported the host up, 99 closed TCP ports and 8080/tcp open (service label: http-proxy).
Process attribution	Listener PID 6; /tools/node/bin/node /datalab/web/app.js; dashboard labels it likely Google Colab-managed infrastructure.
Verdict	Scan execution successful. The finding is not evidence that the user's personal computer or project application is exposed.

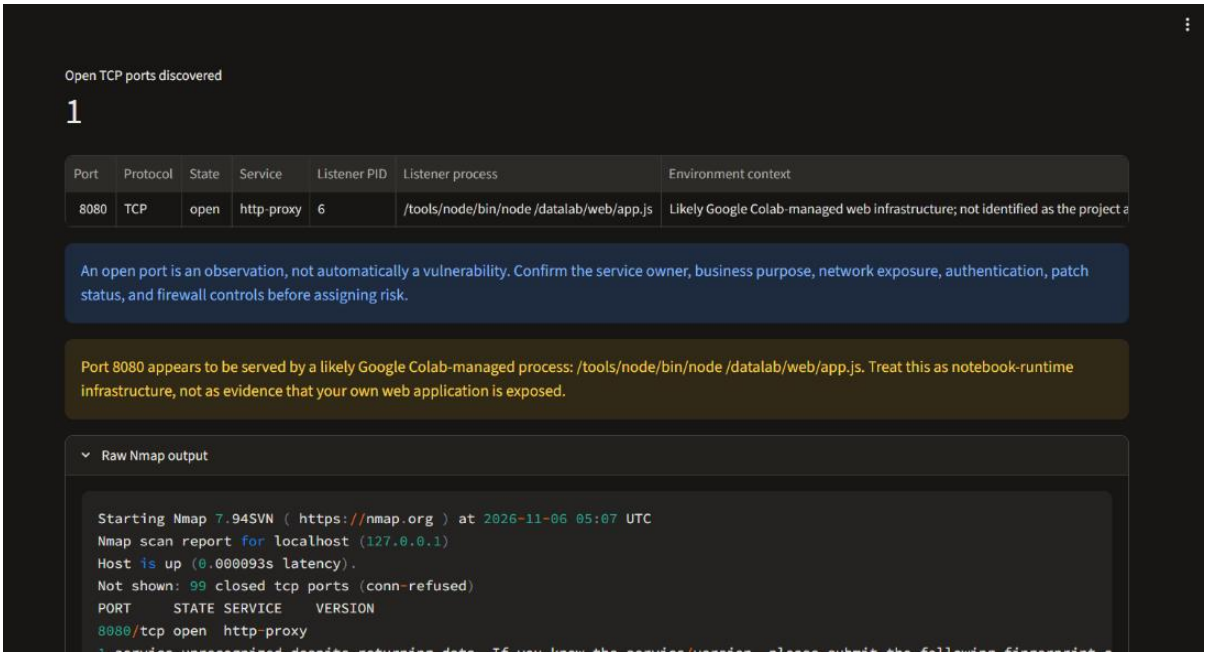
Evidence: Dashboard Screenshots, section “Bounded Nmap Scan”; raw Nmap output is also included in the audit report.

Starting Nmap 7.94SVN (https://nmap.org) at 2026-11-06 05:07 UTC
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000093s latency).
Not shown: 99 closed tcp ports (conn-refused)
PORT STATE SERVICE VERSION
8080/tcp open http-proxy
1 service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint at https://nmap.org/cgi-bin/submit.cgi?new-service :
SF-Port8080-TCP:V=7.94SVN%I=7%D=11/6%Time=6AED60CC%P=x86_64-pc-linux-gnu%r
SF:(GetRequest,52,"HTTP/1\.1\x20404\x20Not\x20Found\r\nDate:\x20Fri,\x2006
SF:\x20Nov\x202026\x2005:08:28\x20GMT\r\nConnection:\x20close\r\n\r\n")%r(

SF:HTTPOptions,52,"HTTP/1\.1\x20404\x20Not\x20Found\r\nDate:\x20Fri,\x2006
SF:\x20Nov\x202026\x2005:08:28\x20GMT\r\nConnection:\x20close\r\n\r\n")%r(
SF:RTSPRequest,52,"HTTP/1\.1\x20404\x20Not\x20Found\r\nDate:\x20Fri,\x2006
SF:\x20Nov\x202026\x2005:08:28\x20GMT\r\nConnection:\x20close\r\n\r\n")%r(
SF:FourOhFourRequest,52,"HTTP/1\.1\x20404\x20Not\x20Found\r\nDate:\x20Fri,
SF:\x2006\x20Nov\x202026\x2005:08:28\x20GMT\r\nConnection:\x20close\r\n\r\
SF:n")%r(Socks5,2F,"HTTP/1\.1\x20400\x20Bad\x20Request\r\nConnection:\x20c
SF:lose\r\n\r\n")%r(Socks4,2F,"HTTP/1\.1\x20400\x20Bad\x20Request\r\nConne
SF:ction:\x20close\r\n\r\n")%r(RPCCheck,2F,"HTTP/1\.1\x20400\x20Bad\x20Req
SF:uest\r\nConnection:\x20close\r\n\r\n")%r(DNSVersionBindReqTCP,2F,"HTTP/
SF:1\.1\x20400\x20Bad\x20Request\r\nConnection:\x20close\r\n\r\n")%r(DNSSt
SF:atusRequestTCP,2F,"HTTP/1\.1\x20400\x20Bad\x20Request\r\nConnection:\x2
SF:0close\r\n\r\n")%r(Help,2F,"HTTP/1\.1\x20400\x20Bad\x20Request\r\nConne
SF:ction:\x20close\r\n\r\n")%r(SSLSessionReq,2F,"HTTP/1\.1\x20400\x20Bad\x
SF:20Request\r\nConnection:\x20close\r\n\r\n")%r(TerminalServerCookie,2F,"
SF:HTTP/1\.1\x20400\x20Bad\x20Request\r\nConnection:\x20close\r\n\r\n")%r(
SF:TLSSessionReq,2F,"HTTP/1\.1\x20400\x20Bad\x20Request\r\nConnection:\x20
SF:close\r\n\r\n")%r(Kerberos,2F,"HTTP/1\.1\x20400\x20Bad\x20Request\r\nCo
SF:nnection:\x20close\r\n\r\n")%r(SMBProgNeg,2F,"HTTP/1\.1\x20400\x20Bad\x
SF:20Request\r\nConnection:\x20close\r\n\r\n")%r(X11Probe,2F,"HTTP/1\.1\x2
SF:0400\x20Bad\x20Request\r\nConnection:\x20close\r\n\r\n")%r(LPDString,2F
SF:,"HTTP/1\.1\x20400\x20Bad\x20Request\r\nConnection:\x20close\r\n\r\n")%
SF:r(LDAPSearchReq,2F,"HTTP/1\.1\x20400\x20Bad\x20Request\r\nConnection:\x
SF:20close\r\n\r\n")%r(LDAPBindReq,2F,"HTTP/1\.1\x20400\x20Bad\x20Request\
SF:r\nConnection:\x20close\r\n\r\n")%r(SIPOptions,2F,"HTTP/1\.1\x20400\x20
SF:Bad\x20Request\r\nConnection:\x20close\r\n\r\n")%r(LANDesk-RC,2F,"HTTP/
SF:1\.1\x20400\x20Bad\x20Request\r\nConnection:\x20close\r\n\r\n");

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 51.55 seconds

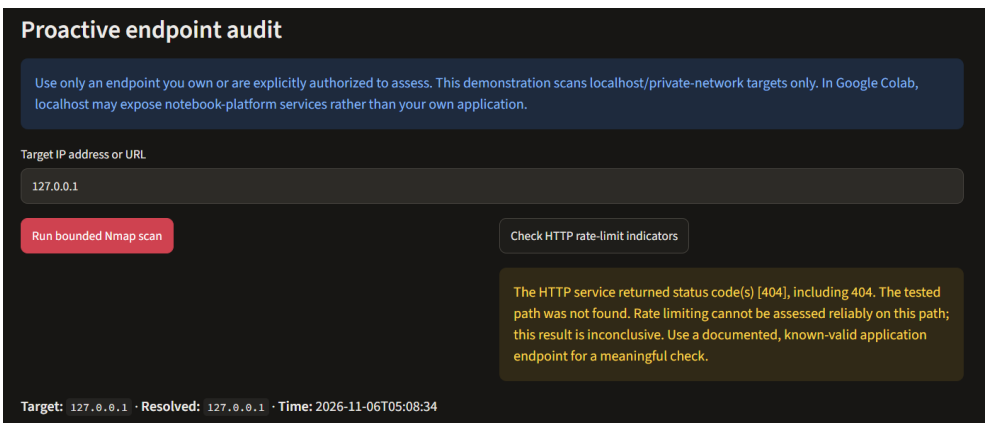


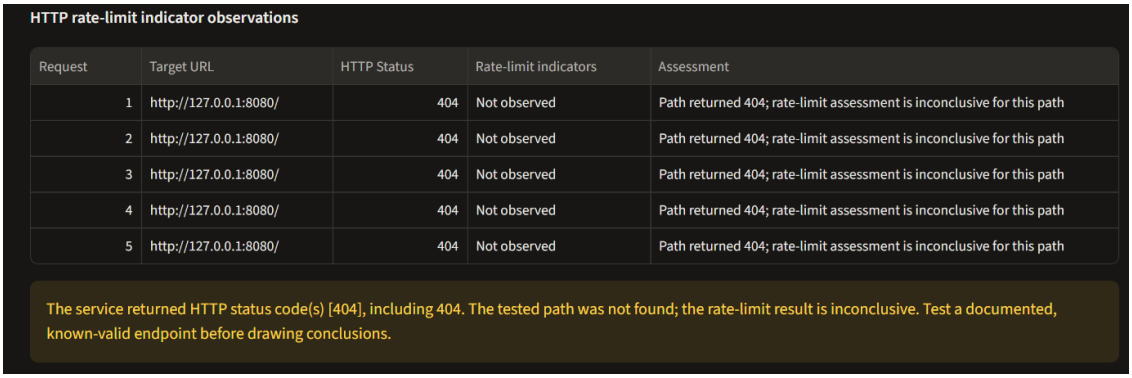


TC02 — HTTP rate-limit indicator check

Field	Details
Objective	Send five sequential HTTP requests and record status codes and common rate-limit headers.
Test input	http://127.0.0.1:8080/
Expected outcome	Record each response and identify explicit throttling responses or standard rate-limit headers where present.
Actual result	Five requests returned HTTP 404. No standard rate-limit indicators were observed.
Verdict	Inconclusive. The tested path was not found; this test does not establish whether rate limiting is enabled or disabled.
Required follow-up	Repeat only against a documented, known-valid endpoint on a system the project team owns or is authorized to test.

Evidence: Dashboard Screenshots, section “HTTP Rate-Limit Indicators”; audit report section 2.





HTTP rate-limit indicator observations

Request	Target URL	HTTP Status	Rate-limit indicators	Assessment
1	http://127.0.0.1:8080/	404	Not observed	Path returned 404; rate-limit assessment is inconclusive for this path
2	http://127.0.0.1:8080/	404	Not observed	Path returned 404; rate-limit assessment is inconclusive for this path
3	http://127.0.0.1:8080/	404	Not observed	Path returned 404; rate-limit assessment is inconclusive for this path
4	http://127.0.0.1:8080/	404	Not observed	Path returned 404; rate-limit assessment is inconclusive for this path
5	http://127.0.0.1:8080/	404	Not observed	Path returned 404; rate-limit assessment is inconclusive for this path

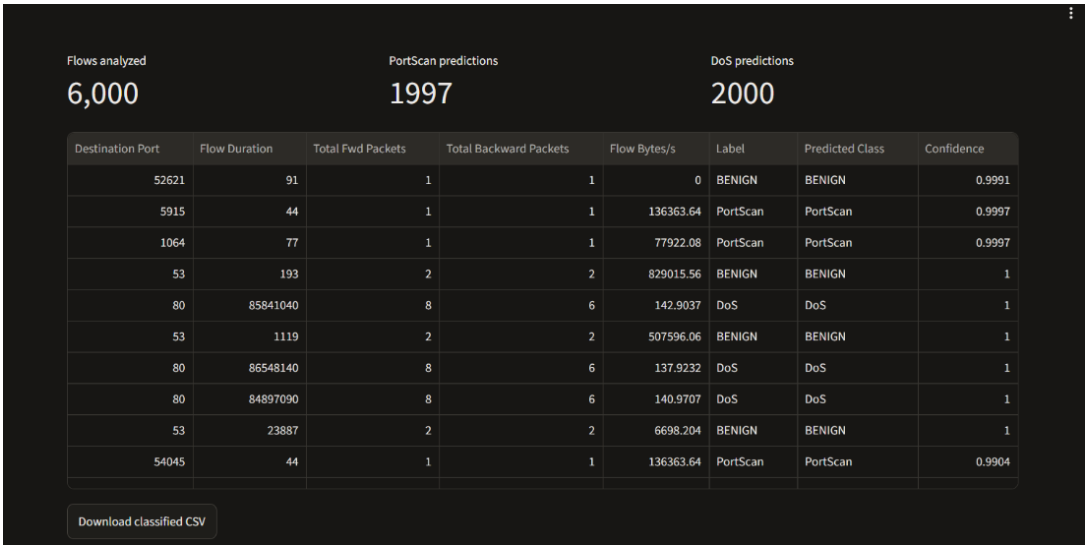
The service returned HTTP status code(s) [404], including 404. The tested path was not found; the rate-limit result is inconclusive. Test a documented, known-valid endpoint before drawing conclusions.

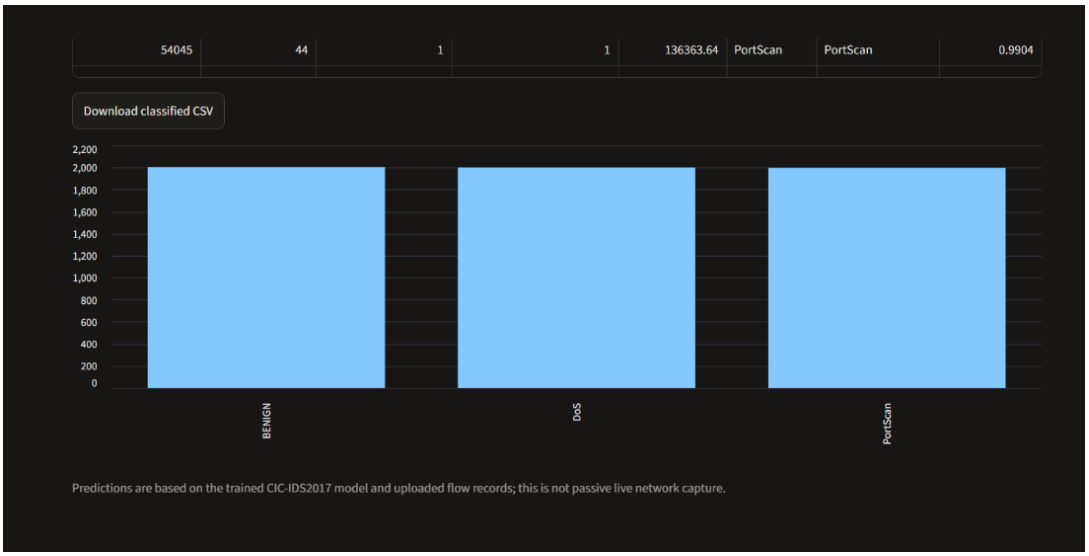
TC03 — AI traffic classification

Field	Details
Objective	Classify uploaded CIC-IDS2017-style flow records using the saved Random Forest model.
Input file	test_flows.csv; 6,000 rows; five model features plus Label.
Expected outcome	Validate required features, predict one of BENIGN / PortScan / DoS for each valid row, and provide confidence values.
Actual result	6,000 records processed. Output file contains the original five features, Label, Predicted Class and Confidence.
Accuracy	5,963 correct of 6,000 labelled records = 99.38%.
Verdict	Successful on the supplied labelled held-out dataset.

Evidence: Dashboard Screenshots, section “AI Threat Detection”; test_flows.csv and classified_network_flows.csv.







Detailed classification metrics

Class	Precision	Recall	F1-score	Support
BENIGN	99.10%	99.25%	99.18%	2,000
PortScan	99.65%	99.50%	99.57%	2,000
DoS	99.40%	99.40%	99.40%	2,000

Confusion matrix (actual rows; predicted columns; order BENIGN, PortScan, DoS):

	BENIGN	PortScan	DoS
BENIGN	1985	7	8
PortScan	6	1990	4
DoS	12	0	1988

Interpretation: The 99.38% score is calculated on the supplied labelled test records. It should not be represented as expected accuracy on live traffic or unrelated network environments. The exported labelled test set is derived from the model's held-out split, so this uploaded-data comparison is not an independent second evaluation.

TC04 — PDF security report generation

Field	Details
Objective	Generate a downloadable PDF consolidating the latest available scan, HTTP and ML results.
Expected outcome	A PDF is created and includes findings, interpretations, limitations and mitigation recommendations.
Actual result	AI_Web_Security_Audit_Report.pdf generated. It includes the port 8080 observation, five HTTP 404 results, class counts, 99.38% holdout and uploaded-labelled accuracy, recommendations and scope note.
Verdict	Successful.

Evidence: AI_Web_Security_Audit_Report.pdf, pages 1–2; Dashboard Screenshot, section “Audit Report Generation”.

